

PV Cell Models and Dynamic Simulation of MPPT Trackers in MATLAB

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Abstract—Technological innovation along with abundance of solar energy in ecosystem has led to its emergence as an alternate, clean and sustainable energy solution. Rising cost and depleting reserves of conventional non-renewable sources has further urged the need to explore and optimize cost of non-conventional energy sources. Intermittent nature of solar energy along with low conversion efficiency of PV (photovoltaic) generators is one of the biggest challenges to its economical and sustainable deployment in power system. Mitigation of these challenges requires research in control algorithms and validated architecture and algorithms on generator and converter side of PV solar generation system. Usage of solar generator before its actual establishment requires accurate and reliable solar cell models to PV generation system further requires MPPT algorithm and architecture to get constant output power and it is one of the most critical component of PV solar generation. This paper compares two existing model with new presented mathematical equation based model to analyse for easy and accurate modelling. PV & VI characteristics of three models are compared for datasheet KC200GT. Further MPPT (Maximum power point tracking) control for PO (perturbation & observation) and IC (incremental conductance) algorithms is developed through MATLAB and compared for different irradiation and temperature levels based on information based on data sheet Warea WS-50 by using proposed solar cell model. The power output of PV array is fed to load through DC-DC Boost converter to verify and validate MPPT control.

Keywords— Photovoltaic array, equivalent model, modelling, MPPT Algorithm, MATLAB SIMULINK

I. INTRODUCTION

In modern era global warming and its implication have become burning issues. Limited reserves of fossil fuels along with their environmental impact has encouraged gradual growth and drift towards green energy sources. Among the available green energy sources photovoltaic (PV) systems are becoming popular due to technological advancement, decreasing cost and acceptability of stakeholders. It has some drawbacks viz. high cost, low efficiency. Although PV power generation is expensive than other resources, government promotions and development in the technology will lead to its competitiveness in future. Recent surveys exhibits that during last few years, solar module's retail price has gone down by 16.95% [1]. In spite of drawbacks it is most promising future energy solution due to its own advantages like abundant, pollution free, distributed throughout the earth and recyclable.

Photovoltaic power generation has no moving parts which reduce its operation and maintenance cost [2].

Two type of PV installations are possible grid connected PV and standalone PV. Efficiency of PV power generation is basically depends on 3 major factors viz. Efficiency of PV panel (8-15%), efficiency of converter (95-98 % [3]) and maximum power point tracking (MPPT) algorithm (over 98% [4]). Improvement in efficiency of the PV panel can be done by up gradation of solar cell material which may increase overall cost. Improvement in MPPT control algorithm is comparatively easier and economic as well. Various MPPT techniques have been reported in the literature [4-10]. MPPT point varies with the variation of temperature and irradiation level; hence it is necessary to track this unique point, MPPT algorithms are to track this maximum point. Moreover irradiation changes rapidly due to changing atmospheric conditions. MPP tracking can be classified in two ways: Mechanical tracking and Electrical tracking [11]. In mechanical tracking PV system is driven mechanically by the stepper motor, while electric tracking based on switching control of power electronic converter. Utilization of solar power is also classified as solar thermal power and solar energy [12].

The output of solar panel is nonlinear in nature and DC; hence for further use it is necessary to convert it in to AC which requires power conditioning unit, as maximum existing loads are of AC load. The solar panel already has low efficiency so additional power conditioning unit will further decrease solar power plant efficiency. Hence efficient and reliable modeling of solar cell is very important and essential. Many solar cell models are already available in literature [13-19] this paper attempt to compare two existing models with the proposed simple mathematical model which can be easily used for any datasheet. Furthermore, two most commonly used MPPT algorithm have been compared and boost converter implemented to verify control algorithm.

II. STRUCTURE OF PV SOLAR CELL AND ARRAY

In photovoltaic energy production process sun light energy is converted to the electricity through photovoltaic effect. It has two layers of semiconductors, coated with photo sensitive material & work similar to p-n junction. When p-n junction absorbs energy of photon it generates charge carriers which get separated at the junction. These charge carriers at junction area create voltage difference and circulate current through the

circuit if it is closed. This current is based on sun light and depends on the irradiation level. If radiation is higher, current contains more photons with energy that is enough to create more electron-hole pairs and resulting more current is produce by cell. The power converted into electricity is given by square of the current multiplied by the resistance of the circuit and rest part will dissipated in the form of temperature to the atmosphere.

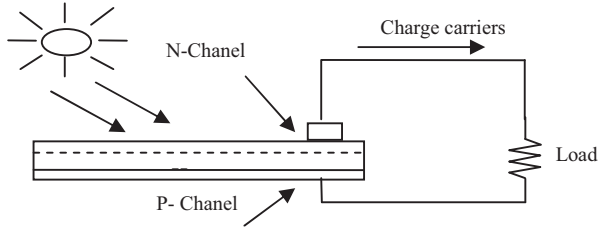


Figure: 1 PV Solar cell

Parallel and series connected solar cells form a solar panel, Parallel and series connected solar cells including bypass diode make solar module, series and parallel connection of solar modules makes solar array. Further the solar panels are connected in series to form solar string. When these solar strings are connected in parallel it makes solar multi string.

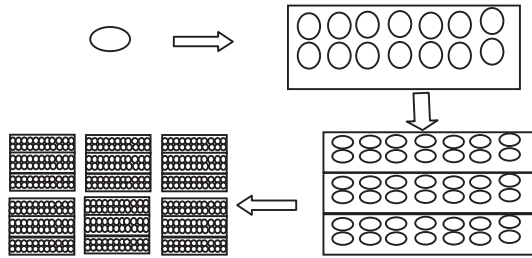


Figure: 2 Photovoltaic cells, modules, array and panels

The power output from PV array, containing N_{ser} in series and N_{par} in parallel PV module can be calculated as:

$$P_{pv} = N_{ser} * N_{par} * FF * V_{oc} I_{sc} \quad (1)$$

Fill factor (FF) is given by following relation:

$$FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}} \quad (2)$$

III. EQUIVALENT CIRCUIT OF SOLAR CELL MODEL

A. Double Diode Model

2-diode model provides better accuracy, 2-diode equivalent circuit model of solar cell depicted in figure3.

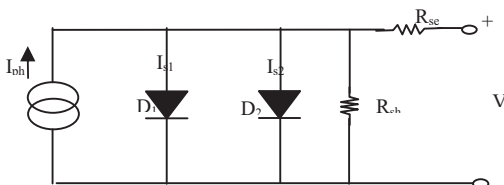


Figure: 3 Double diode models for solar cell

Equation of this model can be expressed as [18]:

$$I = I_{ph} - I_{s1} \left[e^{\frac{q(V + IR_{se})}{kT}} - 1 \right] - I_{s2} \left[e^{\frac{q(V + IR_{se})}{kT}} - 1 \right] - \frac{(V + IR_{se})}{R_{sh}} \quad (3)$$

where I_{ph} is represents photo diode current it is generated by solar cell; I_{s1} is the saturation current of diode D_1 is generated by diffusion mechanism; I_{s2} is the saturation current depends on recombination of charge; R_{sh} is the shunt resistance inversely related to leakage current to the ground; R_{se} is the series resistance signifies internal resistance to current flow & depends upon junction depth impurity and contact resistances; internal losses are represented by R_s & R_{sh} , for an ideal cell R_s should be zero whereas R_{sh} should be infinite, k is Boltzmann's constant (1.38×10^{-23} J/oK), q is the electron charge (1.6×10^{-19} C), T_c is the working temperature of the cell; V is output terminal voltage of solar cell.

B. Single Diode Model

If diode 2 affect is eliminated, the double diode model can be simplified. This assumption is highly acceptable especially under standard test conditions and it introduce very low error o at very low values of irradiation such as $100W/m^2$ [16]. Single diode model is widely used for analysis and designing purposes. The equation of the single diode model is [13]:

$$I = I_{ph} - I_s \left[e^{\frac{q(V + IR_{se})}{kT}} - 1 \right] - \frac{(V + IR_{se})}{R_{sh}} \quad (4)$$

$$I_{s1} = I_{s2} = \frac{I_{sc}}{1000} \left[1 + (T_{cell} - T_{ref})^2 \right] \quad (5)$$

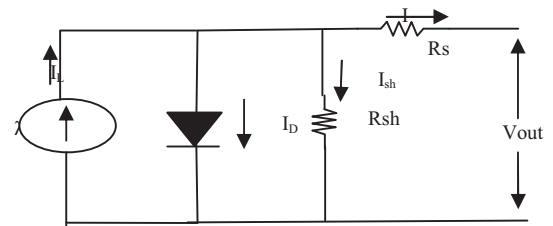


Figure: 4 Single diode model of solar cell

C. PV Solar Array

The photovoltaic array can be simulated with the modified I-V equation (4) [14]. The N_{ser} and N_{par} indexes have to be inserted in the model to make an array from the solar cell.

$$I = I_{ph} N_{ser} - I_s N_{par} \left[e^{\frac{q(V + IR_{se})}{kT}} - 1 \right] - \frac{V + R_s \left(\frac{N_{ser}}{N_{par}} \right) I}{R_{sh} \left(\frac{N_{ser}}{N_{par}} \right)} \quad (6)$$

Where N_{ser} , N_{par} are number of series & parallel cell

respectively.

D. Evaluation of MPPT Algorithm

It is very important to operate PV energy conversion systems at maximum power point (MPP) to obtain the maximum power output from PV generator. Many maximum power point tracking (MPPT) methods have been developed viz. [7].

- (1) Constant Voltage method (CV);
- (2) Perturbation and Observation method (PO);
- (3) Incremental Conductance method (IC) etc.

The CV method is very basic and sets the PV generator current to zero to allow measurement of the PV's open circuit voltage, due to this it suffers from many problems viz. wastage of available energy, interruption of power supply. The last two methods are called "hill-climbing" methods, due to the fact that the PV power-voltage curves having only one peak value over a range of voltage. The main advantages of the PO method are that it can be implemented easily but it has certain limitations i.e. response is slow, oscillations around the MPP in steady state operation. Also, its performance is not very accurate under fast changing atmospheric conditions. The IC method overcomes the limitations of PO method and has higher accuracy and efficiency under fast changing atmospheric conditions. This method is based on Incremental Conductance of the PV module by comparing incremental conductance with instantaneous conductance but this method has also some limitations viz., response time is high and also it is not economic for small-scaled PV plants. Figure 5 and 6 shows the flow chart of PO and IC algorithms respectively.

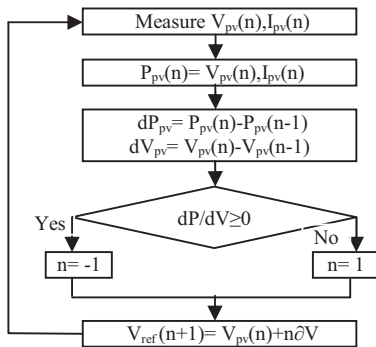


Figure: 5 Control flow chart of P&O algorithm [10]

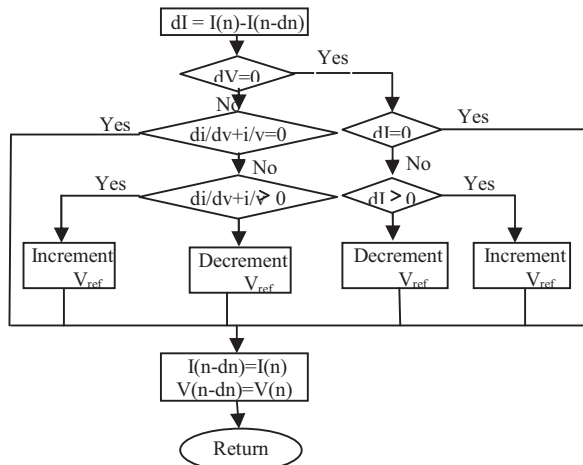


Figure: 6 Control flow chart of IC algorithm [10]

E. Description of Complete Model

In this paper the PO and IC algorithm have been developed to extract maximum power from the array. The change in duty cycle is necessary to control output voltage derived by control algorithm. Maximum power output by the solar panel is provided by MPPT circuit. Boost converter is designed to increase the voltage in steps of the input DC levels.

The PV system is modelled using Power System Block set under Matlab- Simulink based on Datasheet WS-50. It is used to modelling and analysis and considered as model 1. Model 2 is 2-Diode model [18] and model 3[14] is proposed for PV electrical module, comparison have been done for trio models. Figure 7 shows complete block diagram of MPPT implementation.

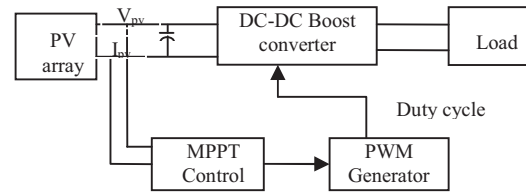


Figure: 7 Block diagram of MPPT implementation

IV. SIMULATION OF PV CELL WITH MPPT CONTROL ON MATLAB/SIMULINK

The details of Datasheet KC200GT are presented in Appendix. This paper developed simple solar cell model based only on single diode mathematical equations converting these equation in Simulink mode by using Simulink block set. This model can be changed by user for any data sheet values just by replacing the information available on manufacturer's data sheet, also can be used for any configuration of array by just multiplying require number of series and parallel cell. Figure 8 represents Simulink setup for PV array.

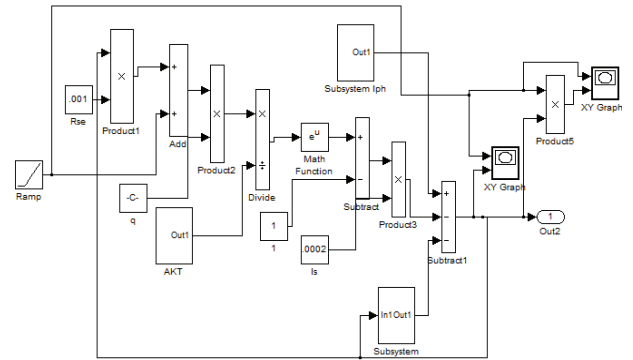


Fig 8 Matlab/Simulink Setup for PV array

The same model as shown in figure 8 is used for the datasheet parameter WS-50 is given in Appendix for MPPT implementation. MPPT control developed in Simulink on the basis of control flow chart, Simulink models of PO and IC algorithms have been shown in Figure 9 & 10.

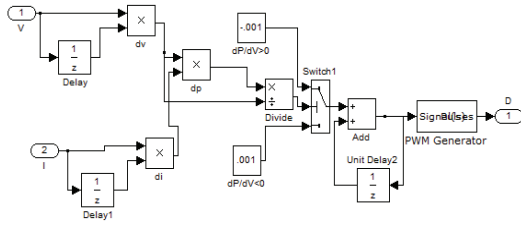


Figure: 9 Matlab/Simulink setup for MPPT PO method

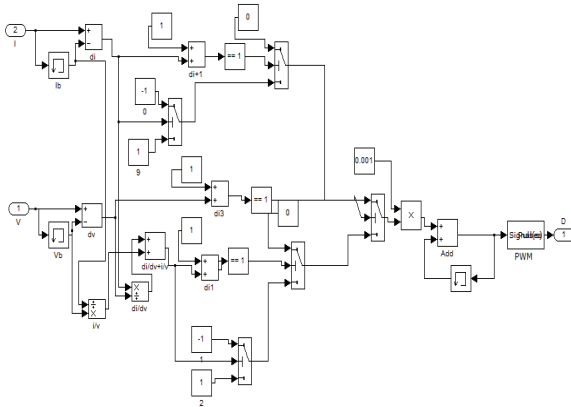
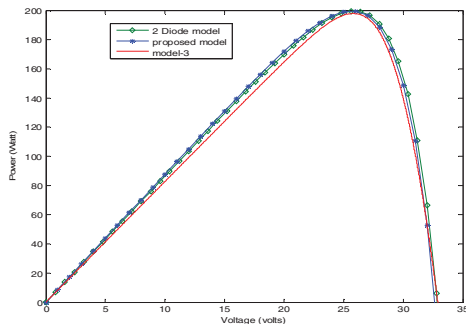


Figure: 10 Matlab/Simulink setup for MPPT IC method

V. RESULTS

V-I and P-V characteristics of simulated trio models at STC has been given in Figure 11 and 12. The curves almost match to each other and give similar result as per data sheet KC200GT. Number of PV cell connected in series or parallel can directly be fed in to model-1 which reduced simulation time, whereas this cannot be done with the model 3. Further 2-diode model (model-2) is based on iterative method again involved long computation time. Hence less simulation time is taken by model1. In terms of accuracy model 2 is highly accurate and model 1 and model 3 having acceptable



accuracy. Model 2 is most complex model whereas model 1

Figure: 11 IV Characteristic of module

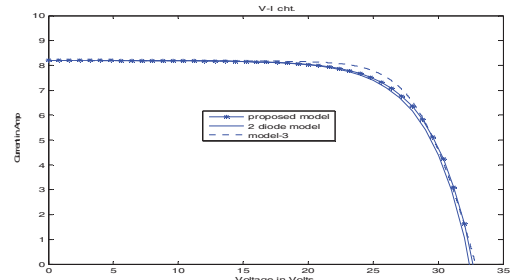


Figure: 12 PV Characteristic of module at STC

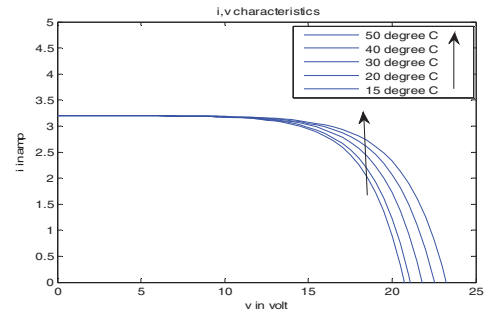


Figure: 13(a) IV characteristics at different temperature (deg C)

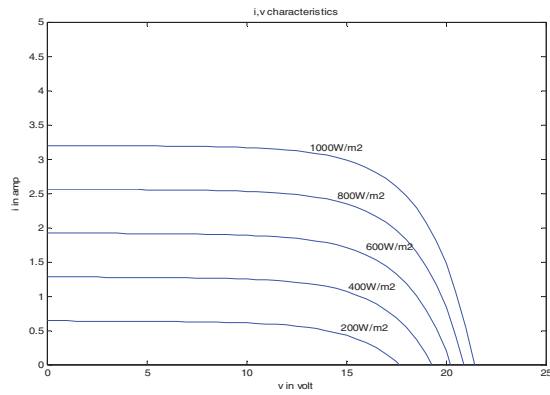


Figure: 13(b) IV characteristics at different irradiation (W/m²)

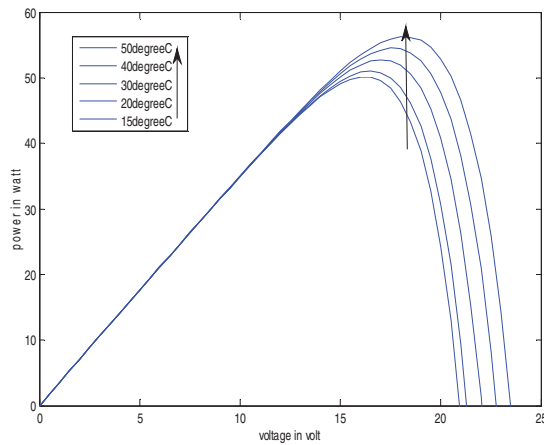


Figure: 14(a) PV characteristics at different temperature (deg C)

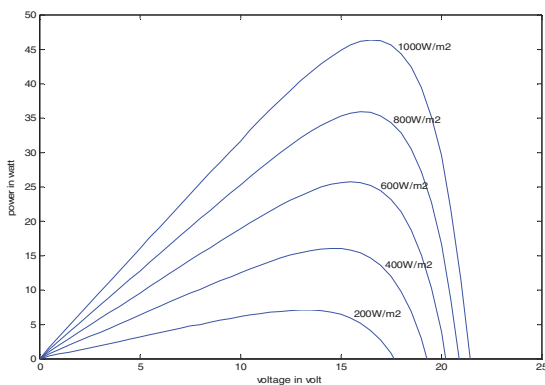


Figure: 14(b) IV characteristics at different irradiation (W/m²)

Figure 15 depicted 9-kW PV array of the sun power modules (WS-50), the array consists of 10 strings of 18 series connected modules connected in parallel ($50 \times 18 \times 10 \text{ W} = 9 \text{ kW}$).

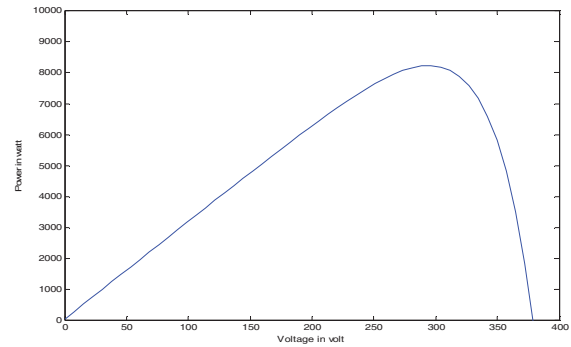


Figure: 15 PV Characteristics of 9 kW PV array

Figure 16 gives MPPT result for the IC and PO algorithm respectively for STC conditions.

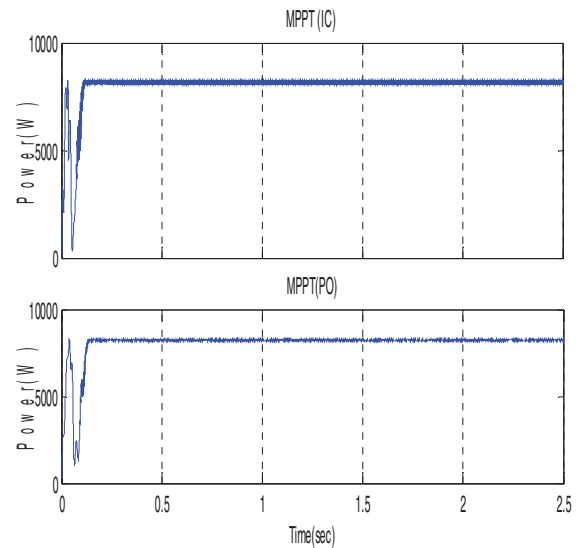


Figure: 16 MPPT for 9Kw PV array through IC & PO control

Figure 17 shows variation in temperature and irradiation levels and corresponding IC and PO responses. The influence of the temperature is slow and both two MPPT methods are easily tracking MPP. It is clearly shown in Figure 17 that for the changes in atmospheric condition, both the algorithms reached at MPP but the dynamic response of PV system for two MPPT algorithms are different for same atmospheric condition. It can be concluded that the obtained power response shows comparatively faster response is provided by the IC to the changes. The IC MPPT is the most adapted for these situations because it reaches the new MPP faster than the PO MPPT whereas PO is easy to implement.

Figure 18 finds differential error between both implemented algorithms.

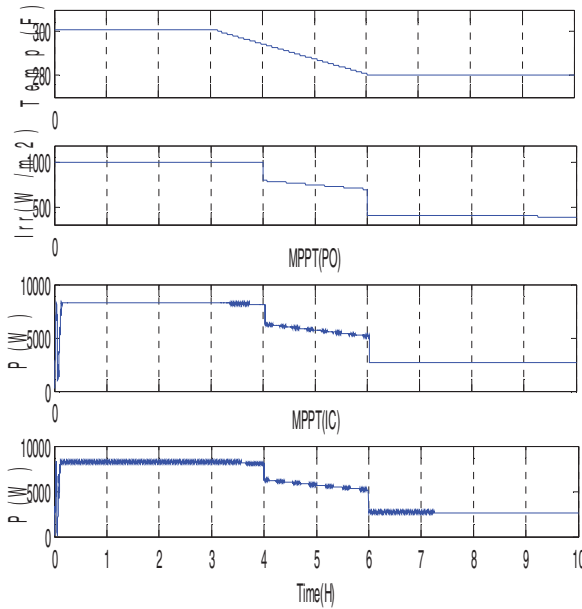


Figure: 17 Different temperature, irradiation levels and corresponding MPPT IC & MPPT PO response

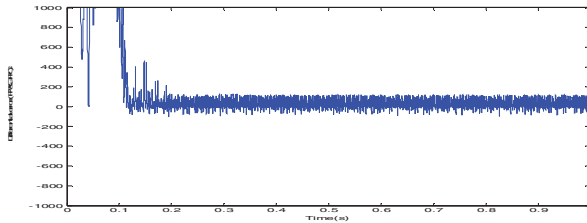


Figure: 18 Differential errors (PPO-PIC)

VI. CONCLUSION

Output characteristic of PV array depends on environmental conditions. Low conversion efficiency of PV requires reliable and optimized modeling / simulation of PV system before its practical establishment. This paper proposes and validates alternate model of PV solar cell which is simple, easy to interface with power conditioning units, requires less simulation time and gives faster dynamic response. Different MPPT algorithm method can further be verified through usage of proposed model. PV generator has nonlinear characteristic and hence MPPT methods are critical to performance of PV generation system. Comparison was done between two commonly used algorithms under different irradiation and temperature level to validate the effectiveness of the model. The IC algorithm for MPPT provides better efficiency for rapidly changed atmospheric conditions whereas PO is less

complex. The paper provides the basic ground work for grid connected and islanded PV generation system.

APPENDIX

Parameters	KC200GT	WS-50
I_{mp}	7.61A	2.94A
V_{mp}	26.3V	17.0V
P_{max}	200.143W	50W
I_{sc}	8.21A	3.17A
V_{oc}	32.9V	21.0 V
K_v	-0.123V/K	0.123 V/K
K_i	0.0032A/K	4.4mA/K

ACKNOWLEDGMENT

The secondary author wishes to thank the Department of Science and Technology for the Sponsored project no. SR/FTP/ETA-20/2010 under the Fast Track Young Scientist Scheme.

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